

# Short-horizon drone RSRP: a failed transferable correction

**P100-038 · Source-aware scientific reference.** Unsupported as an improvement: the development-selected reference fails against the strongest confirmation comparator. It remains a reproducible negative reference; no replacement was selected after exposure.

## Scenario and measured endpoint

First reported RSRP-vector component at first recorded point 1.0–1.6 s after issuance in dBm. At a recorded RSRP sample, use only component 1 in the native comma-separated RSRP vector over the preceding 20 seconds. Predict first subsequently recorded component-1 value at 1.0–1.6 s. First qualifying issuance per target; no actual future horizon feature in predictor. Unknown component antenna/beam semantics prevent propagation-law claims.

Entire flight; forward folds 1->2,1-2->3,1-3->4; flight5 confirmation. Same installation/day only. Development contains 3,595 eligible observations in 4 groups. Confirmation contains 486 eligible observations in 1 groups. Telemetry files1–2 do not overlap the radio clock windows and contain repeated headers; no invented offset or geometry pairing. Vector component ordering is taken literally, not identified as a specific antenna.

## Findings and executable relation

The correction represents noise-dependent mean reversion. Its apparent development benefit did not transfer to the final flight. No new propagation law or operational reliability improvement is supported.

The reference selected before confirmation is attempt-005:

$$\hat{r}_{t+h} = r_t + c (\bar{r}_{5,t} - r_t) \frac{s_{5,t}}{1 \text{ dB} + s_{5,t}}$$

r is the first comma-separated component of the native RSRP vector (dBm); h is the next measured horizon in[1.0,1.6]s, never a prediction input. The trailing 5s mean and population standard deviation s use only available past records. c is dimensionless. The component is not mapped to a known antenna or beam.

| Coefficient | Exact fitted value |
|-------------|--------------------|
| c           | 0.6778972850975561 |

All numerical constants, transformations, alternative equations and fitted coefficients are executable in run.py and serialized in rules.json. Any algebraic consequence of this equation is an implication of the same fitted model, not a second independently validated finding.

## Experimental method and evidence

5 substantive development attempts tested competing representations, controls and ablations. Fits use equal-group weighted least squares or declared robust loss, with all coefficients and radial-basis centers learned only inside each training fold. Selection minimizes mean group MAE, preferring fewer coefficients within 1% of the minimum. Separate confirmation checks the frozen states and performs no fitting. Baselines share permitted information.

Primary errors are in dB; each group contributes equally regardless of row count.

| Frozen model | Development MAE | Confirmation MAE | Worst confirmation group |
|--------------|-----------------|------------------|--------------------------|
| reference    | 1.21588         | 1.04082          | 1.04082                  |
| constant     | 4.48961         | 4.46257          | 4.46257                  |
| flexible     | 1.20931         | 0.928604         | 0.928604                 |

| Frozen model | Development MAE | Confirmation MAE | Worst confirmation group |
|--------------|-----------------|------------------|--------------------------|
| persistence  | 1.25461         | 0.944856         | 0.944856                 |
| attempt-001  | 1.25773         | 0.99601          | 0.99601                  |
| attempt-002  | 1.2287          | 1.04244          | 1.04244                  |
| attempt-003  | 1.21577         | 1.00498          | 1.00498                  |
| attempt-004  | 1.21068         | 0.987291         | 0.987291                 |
| attempt-005  | 1.21588         | 1.04082          | 1.04082                  |

The frozen reference changes confirmation MAE by -12.1% relative to the strongest evaluated baseline `flexible` (0.928604 dB); it wins on 0/1 whole groups. This is descriptive, not a population significance test. Rows within runs, days, weeks, profiles or casts are dependent. The complete group errors, bias, RMSE and 90th-percentile errors are in `evidence/by_group.csv`.

### Falsification, limits and value

- Trend extrapolation is not a reliable improvement over persistence in this cohort.
- The source telemetry clocks for flights1–2 do not overlap radio windows. This is an input audit limitation, not a discovered physical relationship.
- Report whole-group errors and ranges; no independent-row population CI.
- Temporal/spatial dependence and small number of sites/floats limit inference.
- Public data and known physical parameterizations preclude automatic novelty certification. Predictive accuracy does not establish an industrial intervention or a new physical law.

Schema semantics come before geometric modeling: clock-incompatible telemetry and unlabelled vector entries do not justify altitude or antenna claims. Preserve failed transfer even when development scores improve.

### Reproduction and future evaluation

Run `python run.py` in this folder to verify allowlisted hashes and reproduce every saved confirmation prediction. This executes trusted bundled reference code; scoring an external prediction file should use the shared Principia evaluator without executing submitted code. Numerical alternatives need not resemble this equation. Use the exact registered task, timing/input budget, units, eligibility and group weighting. New endpoints or information access require a separately reviewed task version.

All confirmation outcomes are now exposed to future users. Source-stage exposure is recorded in `task_spec.json`; the first-use internal freeze history is preserved in research history. Independent source data are needed for a new confirmation campaign.

### Primary sources and existing analyses

- <https://zenodo.org/records/16673883> — Dataset creator describes five drone measurement iterations from one private5G network. Native vector component identities and mismatched earlier telemetry windows are unresolved.

[https://www.etsi.org/deliver/etsi\\_ts/138200\\_138299/138215/17.04.00\\_60/ts\\_138215v170400p.pdf](https://www.etsi.org/deliver/etsi_ts/138200_138299/138215/17.04.00_60/ts_138215v170400p.pdf) — 3GPP/ETSI physical-layer RSRP measurement definition; provides quantity context, not identification of each Nemo export vector component.